

Solutions for Problem Set 4

Mathematics for Social Scientists

EXERCISE 1. Determine whether the following functions are convex or concave (if any at all) over the domain that they exist:

i. $f(x) = x^4$

ii. $g(x) = 8x + 26$

iii. $h(x) = \sqrt{x}$

iv. $p(x) = -x^{131894}$

v. $r(x) = e^x$

vi. $s(x) = \log(x)$

i. convex

ii. linear (both concave and convex)

iii. concave

iv. concave

v. convex

vi. concave

EXERCISE 2. For the following functions, determine **the interval(s)** over which the function is convex or concave (if at all), and identify any saddle (inflection) points:

i. $f(x) = \frac{1}{x}$

ii. $f(x) = \frac{x^{1-\gamma}}{1-\gamma}$ for $0 < \gamma < 1$

iii. $f(x) = xe^x$

iv. $f(x) = \sin(x)$

v. $f(x) = \lambda e^{-\lambda x}$ for $\lambda > 0$

Remark: This is the exponential distribution

vi. $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Remarks: This is the standard normal distribution

vii. $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

Remarks: This is the general normal distribution

- i. $(-\infty, 0)$: concave
 $(0, \infty)$: convex
 No saddle point in domain

- ii. $(0, \infty)$: concave
 Domain: $x > 0$; no saddle point

- iii. $(-\infty, -2)$: concave
 $(-2, \infty)$: convex
 Saddle pt = -2

- iv. Convex on $(\pi + 2\pi k, 2\pi + 2\pi k)$,
 Concave on $(2\pi k, \pi + 2\pi k)$,
 Saddle pts: $\pi k, \forall k \in \mathbb{Z}$

- v. Convex

- vi. Convex on $(-\infty, -1] \cup [1, \infty)$
 Concave on $(-1, 1)$
 Saddle pts: $\{-1, 1\}$

- vii. Concave on $(\mu - \sigma, \mu + \sigma)$
 Convex on $(-\infty, \mu - \sigma] \cup [\mu + \sigma, \infty)$
 Saddle points: $\{\mu - \sigma, \mu + \sigma\}$

EXERCISE 3. Find the zeros and local maxima/minima of the following functions.

(Recall: the zeros of f are x such that $f(x) = 0$, whereas the local maxima/minima are x such that $f'(x) = 0$)

i. $g : [-1, 1] \rightarrow \mathbb{R} : x \mapsto x^2$

ii. $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto x \ln(x)$

iii. $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto x^2 \ln(x)$

- i. zero at $x = 0$; local minimum at $x = 0$
- ii. zero at $x = 1$; local minimum at $x = e^{-1}$
- iii. zero at $x = 1$; local minimum at $x = e^{-1/2}$

EXERCISE 4. For the following functions f ,

- (a) differentiate f
- (b) find the local maxima and minima by solving $f'(x) = 0$
- (c) test whether these points are minima or maxima by finding $f''(x)$
- (d) calculate the value of f at these points
- (i) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 2x^3 - 9x^2 - 24x + 8$
- (ii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 3x^4 - 4x^3$
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 10xe^{-\frac{x^2}{2}}$
- (iv) $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto -10 \cdot \frac{\log(x)}{x^2}$
- (v) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto (x^4 - x^2)^2$

(i) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 2x^3 - 9x^2 - 24x + 8$

(a) $f'(x) = 6(x - 4)(x + 1)$

(b) local extrema = $\{-1, 4\}$

(c) -1 is a local max, 4 is a local min

(d) $f(-1) = 21$, $f(4) = -104$

(ii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 3x^4 - 4x^3$

(a) $f'(x) = 12x^2(x - 1)$

(b) local extrema = $\{0, 1\}$

(c) 1 is a local min, 0 saddle pt

(d) $f(0) = 0$, $f(1) = -1$

(iii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 10xe^{-\frac{x^2}{2}}$

(a) $f'(x) = 10e^{-\frac{x^2}{2}}(1 - x^2)$

(b) local extrema = $\{-1, 1\}$

(c) -1 is a local min, 1 is a local max

(d) $f(-1) = -10e^{-1/2}$, $f(1) = 10e^{-1/2}$

(iv) $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto -10 \cdot \frac{\log(x)}{x^2}$

(a) $f'(x) = -10 \frac{1 - 2\log(x)}{x^3}$

(b) local extrema = $\{e^{1/2}\}$

(c) $e^{1/2}$ is a local min

(d) $f(e^{1/2}) = -\frac{5}{e}$

(v) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto (x^4 - x^2)^2$

(a) $f'(x) = 4x^3(x-1)(x+1)(2x^2-1) = 4x^3(x-1)(x+1)(\sqrt{2}x-1)(\sqrt{2}x+1)$

(b) local extrema = $\left\{-1, 0, 1, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\}$

(c) $-1, 0, 1$ are local min; $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ are local max

(d) $f(-1) = f(0) = f(1) = 0$, $f\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{16} = f\left(-\frac{1}{\sqrt{2}}\right)$

EXERCISE 5. Consider the function $f(x) = (x^2 + 2x)e^{-x}$.

- (i) Determine for which values of x the value of the function is equal to zero.
- (ii) Calculate $f'(x)$ and find the extreme points of f . What is the maximum/the minimum of the function?
- (iii) Does the function have inflection points?
- (iv) Sketch the function and specify whether it is convex/concave (in sections).

Solution:

- (i) Determine for which values of x the value of the function is equal to zero.

We have to set $(x^2 + 2x)e^{-x} = 0$. We know that $e^{-x} > 0$ for all $x \in \mathbb{R}$. Thus,

$$x^2 + 2x = 0 \implies x = \frac{-2 \pm \sqrt{4}}{2} = -1 \pm 1.$$

The roots of the function are $x = -2$ and $x = 0$.

- (ii) Calculate $f'(x)$ and find the extreme points of f . What is the maximum/the minimum of the function?

$$\begin{aligned} f(x) &= (x^2 + 2x)e^{-x}, \\ f'(x) &= -(x^2 + 2x)e^{-x} + (2x + 2)e^{-x} = e^{-x}(-x^2 + 2), \\ f''(x) &= (x^2 + 2x)e^{-x} - 2(2x + 2)e^{-x} + 2e^{-x} = (x^2 - 2x - 2)e^{-x}. \end{aligned}$$

We take the FOC $f'(x) = 0$ to look for critical points:

$$e^{-x}(-x^2 + 2) = 0 \implies -x^2 + 2 = 0 \implies x = \pm\sqrt{2}.$$

Check the SOC with $f''(x) = (x^2 - 2x - 2)e^{-x}$ (and $e^{-x} > 0$):

$$f''(-\sqrt{2}) > 0 \implies \text{local minimum at } x = -\sqrt{2}, \quad f''(\sqrt{2}) < 0 \implies \text{local maximum at } x = \sqrt{2}.$$

As the domain is not limited, consider the limits to decide on global behavior:

$$\lim_{x \rightarrow -\infty} (x^2 + 2x)e^{-x} = +\infty, \quad \lim_{x \rightarrow +\infty} (x^2 + 2x)e^{-x} = 0.$$

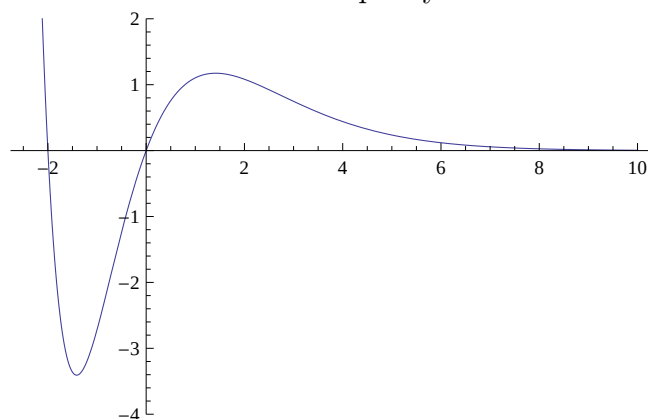
Therefore, the function **does not have a global maximum** and **has a global minimum at $x = -\sqrt{2}$** .

- (iii) Does the function have inflection points?

Yes. Set $f''(x) = 0$:

$$(x^2 - 2x - 2)e^{-x} = 0 \implies x^2 - 2x - 2 = 0 \implies x = 1 \pm \sqrt{3}.$$

- (iv) Sketch the function and specify whether it is convex/concave (in sections).



The function is neither convex nor concave as a whole. From $f''(x)$ we see it is **convex on $(-\infty, 1 - \sqrt{3}) \cup [1 + \sqrt{3}, \infty)$** and **concave on $[1 - \sqrt{3}, 1 + \sqrt{3}]$** .